

● FYDP-I · SEPTEMBER 2026

FROM REPOSITORY EVIDENCE TO **CLOUD** INFRASTRUCTURE ●

Depot reads a repo, builds one model, provisions on **AWS, GCP** or **Azure**.

REPOSITORY-FIRST

EVIDENCE-AWARE

APPROVAL-GATED

ORDINARY REPO

—>

**APPLICATION
MODEL**

—>

AWS

GCP

AZURE

BUILDING IS FAST NOW. SHIPPING IS STILL **MANUAL WORK**●

01**WHERE KNOWLEDGE SITS**

Scattered across code, docs and tooling.

02**WHAT SMALL TEAMS FACE**

No dedicated platform engineer.

TERRAFORM ALONE ISN'T SAFE.

PROPOSAL §1

THE GAP, AND WHAT DEPOT DOES ABOUT IT

Existing tools host the app, force rewrites, or generate unsafe infra. Depot reads evidence, builds one model, provisions only after an approved plan.

EVIDENCE FIRST

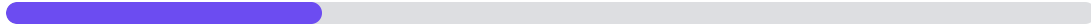
REVIEWED PATTERNS ONLY

HUMAN APPROVAL FIRST

THE NEED IS LARGER THAN THE EXPERTISE

REPO WORK

Developers on GitHub **180M+**



Repositories on GitHub **630M**



GitHub Octoverse 2025

INFRASTRUCTURE-AS-CODE DIFFICULTY

Stack Overflow questions **52,692**



Answers **64,078**



Begoug et al. [6]

FOUR CATEGORIES. EACH SOLVES **PART** OF IT

LEGEND
✓ yes · ~ partial · — no

APPROACH	YOUR CLOUD	ONE SPEC, THREE CLOUDS	APP MODEL	APPROVAL FIRST	NO REWRITE
MANAGED PLATFORMS	—	—	—	—	✓
BRING-YOUR-OWN-CLOUD	✓	~	—	—	~
FRAMEWORK-DRIVEN	~	~	~	—	—
AI & ENTERPRISE IAC	✓	✓	~	—	✓
DEPOT (PROPOSED)	✓	✓	✓	✓	✓

DEPOT CHECKS EVERY COLUMN.

TABLE 1

SIX THINGS TO BUILD

01**DEFINE THE CONTRACTS**

Versioned schemas with provenance.

02**BUILD THE ANALYZERS**

Every finding evidenced.

03**DESIGN GUIDED INTAKE**

Ask what the repo cannot.

04**CONSTRAIN THE REASONING**

Patterns with alternatives and cost.

05**PROVIDER ADAPTERS**

Terraform adapters for AWS, GCP, Azure.

06**GATE, THEN MEASURE**

Five gates before any change.

BUILT FOR TEAMS WITHOUT A PLATFORM ENGINEER



SMALL TEAMS

Owns account; approves changes.



DEVELOPERS

Reviews model, evidence and plan.



EXPERIENCED ENGINEERS

Engineering view: schemas, mappings, plans.

SIX INSPECTABLE STEPS



STATE FEEDS THE NEXT CHANGE

AI INTERPRETS; SCHEMAS AND APPROVAL EXECUTE.

DETERMINISTIC

WHAT THE SYSTEM ACTUALLY DOES

01**REPOSITORY INTELLIGENCE**

Least-privilege access; findings carry evidence.

02**GUIDED INTAKE**

Traffic, budget, region, compliance.

03**APPLICATION MODEL**

Versioned contract; provenance per field.

04**ARCHITECTURE REASONING**

Reviewed patterns with rationale and cost.

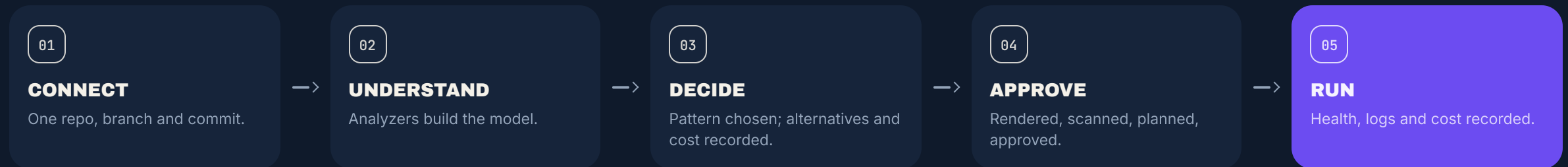
05**APPROVAL GATE**

Nothing applies without approval.

06**DELIVERY AND OPERATIONS**

Commit-pinned CI/CD, logs, health checks.

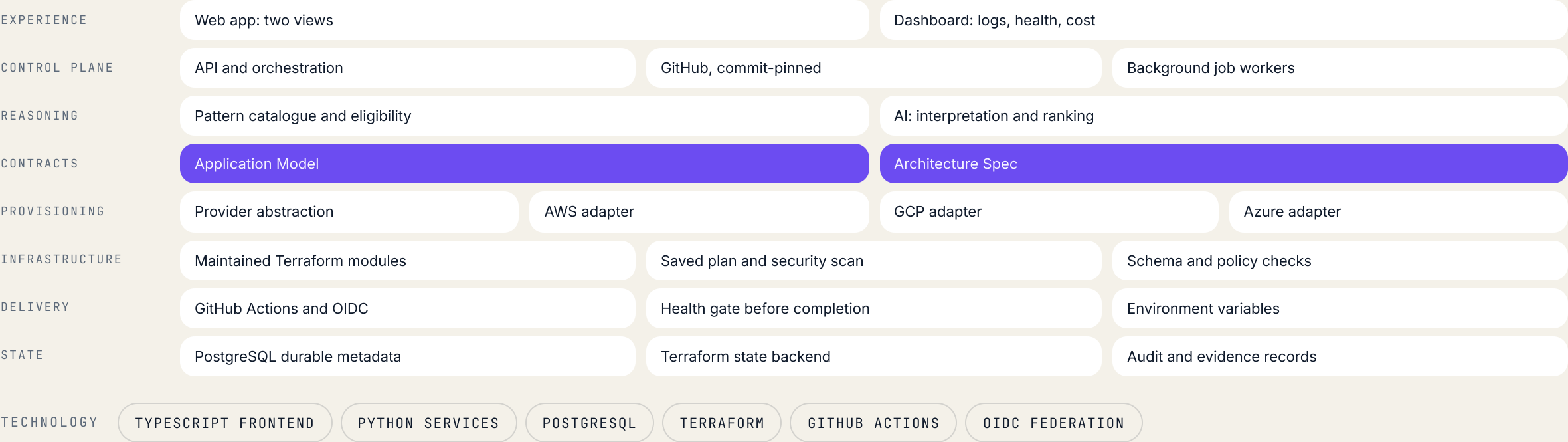
ONE REPOSITORY, END TO END



EVERY STEP LEAVES AN ARTEFACT.

COMMIT TO CLOUD

ONE MODEL BETWEEN EVIDENCE AND EXECUTION



IN SCOPE, AND DELIBERATELY OUT

AREA	IN SCOPE	OUT OF SCOPE
REPOSITORY INPUT	Repo + intake, no rewrite	Rewrite or SDK lock-in
ARCHITECTURE	Web, API, worker; single region	Kubernetes, multi-region active-active
PROVIDERS	AWS, GCP and Azure, one specification	Other providers; forced equivalence
IAC	Reviewed Terraform modules	Free-form generation
ACCOUNTS AND KEYS	Own account; short-lived identity	Managed hosting; stored keys
CHANGES	Plan, approval, audit record	Unapproved destructive operations
COMPLIANCE AND ACCESS	Unsupported cases escalated	Certification; multi-tenant RBAC

UNSUPPORTED CASES ARE ESCALATED.

PROPOSAL §4.6

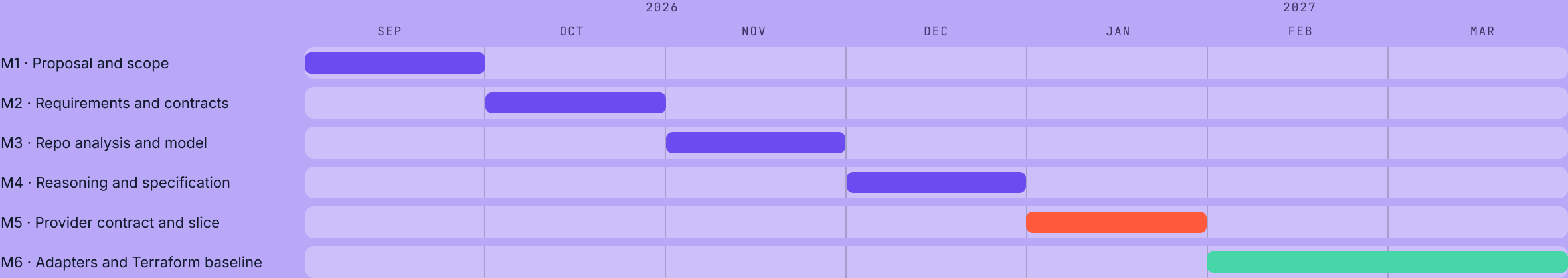
HOW WE’LL PROVE IT WORKS

DEPOT SHOULD...	CHECK	PASS
ACCURATE REPRESENTATION	Precision, recall, F1	Schema-valid, provenance
SUITABLE ARCHITECTURE	Blind top-1 review	Accepted by blind reviewer
VALID, SAFE INFRA	Schema, validate, plan pass rates	All pass; zero high-severity
DEPLOY AND RECOVER	Six scenario-provider pairs	Six pairs healthy
COST ESTIMATE	Error vs provider estimate	Error reported and explained
TARGET USERS	Task completion, errors, SUS	No critical errors; SUS kept

FAILURES ARE CLASSIFIED BY STAGE — NEVER CONFLATED.

TABLE 5

PROPOSAL TO A CROSS-CLOUD SLICE



THREE OWNERS. SHARED REVIEW.

AR**AHSAN RIAZ**

479561

Repo analysis, model, evaluation metrics.

MW**MUHAMMAD WALEED**

455706

Frontend, dashboard, operations UI.

MFA**MUHAMMAD FAIZAN ANWAR**

455259

Reasoning, provisioning, three provider adapters.

NOBODY REVIEWS ONLY THEIR OWN WORK.

HIRA ANWAR · AYESHA HAKIM

WHAT COULD GO WRONG, AND OUR FIX

MULTI-CLOUD EXCEEDS CAPACITY

Small pattern set; features frozen early.

**PROVIDERS NOT BEHAVIOURALLY
EQUIVALENT**

Explicit limits; providers tested separately.

**EVIDENCE INCOMPLETE OR
MISLEADING**

Provenance recorded; unsupported cases rejected.

**AI REASONING IS PLAUSIBLE BUT
WRONG**

Patterns, checks, approval.

SECURITY OR COST EXPOSURE

Least privilege, scanned plans, budget limits.

TERRAFORM OR PROVIDER DRIFT

Pinned versions and automated adapter tests.

BIASED, TOO-SIMPLE CORPUS

Stratified, published selection criteria.

TWO GOALS WE CONTRIBUTE TO



SDG 8 - DECENT WORK

Provider-grade infra without a platform team.



SDG 9 - INNOVATION

Constrained AI: reviewable, validated, human-approved.

MORE PEOPLE SHIPPING SAFELY; ACCOUNTABLE AI.

UN ICONS, UN.ORG

SOURCES & FURTHER READING

[1] GitHub, Octoverse 2025.

[2] Begoug et al., ESEM 2023.

[3] Guerriero et al., ICSME 2019.

[4] Rahman et al., ICSE 2019.

[5] Davidson et al., arXiv:2509.05303, 2025.

[6] Nekrasov et al., arXiv:2512.14792, 2025.

[7] Vargas et al., arXiv:2608.02672, 2026.

[8] Artac et al., ICSE Companion 2017.

[9] Tankov et al., arXiv:2108.07842, 2021.

[10] Provider and identity docs: HashiCorp, GitHub, GCP, Microsoft, AWS, 2026.

[11] Railway, Render, Vercel et al. docs, 2026.

[12] NUST SEECS, FYP Guidelines v1.4, 2017.

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THANK YOU. ANY QUESTIONS?

3

provider paths

20

labelled repositories

0

credentials stored

EVIDENCE-PRESERVING

PROVIDER-INDEPENDENT

APPROVAL-GATED